

Notes: Garin & Silvério (RES, 2024)
How Responsive Are Wages to Firm-Specific Changes in Labour Demand?
Evidence from Idiosyncratic Export Demand Shocks

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1 Big picture

- **Question.** Do firms adjust wages because their own demand changes, or because rival employers in the labour market also change wage competition?
- **Setting.** Portuguese exporters during the 2008–09 Great Recession.
- **Key decomposition.** Firm-level export demand shocks are decomposed into a common product-market component and an idiosyncratic firm component.
- **Main firm result.** Both components affect own-firm output, but only the common component spills over to close competitors.
- **Main worker result.** Idiosyncratic firm demand shocks pass through to incumbent wages even when rival employers are not affected.
- **Magnitude.** A 10% idiosyncratic decline in firm sales reduces incumbent wage growth by about 1–1.5%.
- **Mechanism.** Firm-specific pass-through is evidence of noncompetitive quasi-rents tied to durable employment relationships.

2 Data and measurement

The paper combines Portuguese firm-level administrative data, linked employer–employee records, and international trade data to study how export demand shocks affect firms and workers. The central empirical object is a firm-level foreign demand shock constructed from each firm’s pre-recession export portfolio. The key innovation is that this shock is decomposed into a *common* component, which affects many firms producing the same products, and an *idiosyncratic* component, which is specific to a firm’s pre-recession destination mix.

2.1 Firm sample

The main analysis sample contains 4,173 small and medium exporters. These firms satisfy three main conditions:

- they exported during the pre-recession period 2005–2007;
- they had no more than 100 employees;
- they can be matched to Portuguese firm-level and worker-level administrative records.

The small-and-medium-firm restriction matters because these firms are more likely to be price takers in foreign markets and less likely to affect foreign import-demand measures themselves. The export requirement matters because the paper’s identifying variation comes from changes in foreign demand for the products and destinations that firms served before the Great Recession.

Interpretation: The firm sample is designed to focus on firms that are exposed to foreign demand but are small enough that their own export decisions are unlikely to move foreign market demand. This helps make the export-demand shock plausibly external to the firm.

2.2 Timing of the sample

The empirical design separates the sample into three timing blocks:

- **Pre-period:** 2005–2007. This period is used to measure firms’ export portfolios, product shares, destination shares, baseline characteristics, and worker-firm matches.

- **Transition year:** 2008. This year is often treated separately because the recession begins during the year and adjustment may be partial.
- **Post-period:** 2009–2011. These years capture firm and worker outcomes after the recession-induced foreign demand shock.

A simplified before-after structure is:

$$Pre = \{2005, 2006, 2007\}, \quad Post = \{2009, 2010, 2011\}.$$

Interpretation: Using the pre-period to construct exposure avoids contaminating the shock with firms’ post-recession choices. The post-period is then used to measure how firms and workers respond to demand changes that were largely outside their control.

2.3 Linked employer–employee data

The paper tracks workers employed by the sample firms in 2007. Let $j(i, 2007)$ denote worker i ’s employer in 2007. The worker-level data allow the authors to follow these workers over time and measure whether their wages, employment status, and attachment to the original firm change after the shock.

The key worker groups are:

- **All 2007 workers:** workers employed at sample firms in the baseline year.
- **Attached incumbents:** workers who remain employed at their 2007 employer.
- **Separated workers:** workers who leave the 2007 employer.
- **Workers employed elsewhere:** workers who appear at another employer after leaving.

Interpretation: The worker panel is crucial because it allows the paper to distinguish wage effects for incumbent workers from changes in the composition of the workforce. If a firm-level demand shock raises wages for attached incumbents, the effect is more naturally interpreted as rent sharing within ongoing employment relationships.

2.4 Firm-level outcomes

The paper studies several firm outcomes. The main firm-level variables include:

- **Exports:** the firm’s foreign sales.
- **Sales:** total firm revenue.
- **Output or value added:** production-side measures of firm performance.
- **Payroll:** total wage payments by the firm.
- **Employment:** total number of workers employed by the firm.
- **Hiring and separations:** worker-flow outcomes that show whether adjustment occurs through expansion, contraction, or worker turnover.

A generic firm-level outcome is denoted:

$$Y_{jt},$$

where j indexes firms and t indexes years.

Interpretation: Firm outcomes are used to verify that the foreign demand shock actually moves firm activity. This is the first-stage logic: a demand shock should affect exports, sales, and output before it can plausibly affect wages.

2.5 Worker-level outcomes

The main worker-level outcomes include:

- annual wages;
- wage growth;
- employment status;
- whether the worker remains with the 2007 employer;
- whether the worker moves to another firm;
- whether the worker exits the matched employer–employee data.

A generic worker wage outcome is written as:

$$w_{it},$$

where i indexes workers and t indexes years.

For worker-level regressions, the relevant firm shock is assigned using the worker’s 2007 employer:

$$\Delta\tau_{j(i,2007)}.$$

Interpretation: The assignment of shocks through the 2007 employer is important. It fixes worker exposure before the recession and prevents post-shock job mobility from mechanically changing the treatment variable.

2.6 Export portfolio shares

The demand shock is based on each firm’s pre-recession export portfolio. Let m index products and c index destination countries. The pre-period export share of firm j in product-destination market (m, c) is:

$$s_{jmc}^{pre} = \frac{Exports_{jmc}^{2005-2007}}{\sum_{\mu \in M} \sum_{\kappa \in C} Exports_{j\mu\kappa}^{2005-2007}}.$$

This share measures how exposed firm j was to product m in destination c before the Great Recession.

The firm-level pre-period product share is:

$$s_{jm}^{pre} = \sum_{c \in C} s_{jmc}^{pre}.$$

Interpretation: The export shares are predetermined exposure weights. A firm that exported heavily to a destination-product market before the recession receives more exposure to that market’s later foreign demand change.

2.7 Foreign import-demand changes

For each product-destination market (m, c) , the paper measures the change in foreign import demand using non-Portuguese imports. Denote this external demand measure by:

$$NPI_{mc},$$

where NPI stands for non-Portuguese imports.

A symmetric growth rate is used:

$$\Delta M_{mc} = \frac{NPI_{mc}^{post} - NPI_{mc}^{pre}}{\frac{1}{2} (NPI_{mc}^{post} + NPI_{mc}^{pre})}.$$

This growth-rate formula is useful because it treats increases and decreases symmetrically and is less sensitive to scale than a simple percentage change.

Interpretation: The paper uses non-Portuguese imports so that the shock is not mechanically driven by the Portuguese firms being studied. The demand shifter comes from foreign markets, not from the firm's own realized exports.

2.8 Total firm-level export demand shock

The total export demand shock for firm j is the weighted average of product-destination import demand changes, using the firm's pre-recession export shares:

$$\Delta \tau_j^{tot} = \sum_{m \in M} \sum_{c \in C} s_{jmc}^{pre} \Delta M_{mc}.$$

This is a shift-share measure. The shares are fixed before the shock, and the shifts are foreign import-demand changes during the recession.

Interpretation: $\Delta \tau_j^{tot}$ asks: given the markets firm j served before the recession, how much did foreign demand for those markets change during the recession? It is a predicted demand shock, not the firm's realized export growth.

2.9 Why the total shock is decomposed

The total shock can mix two different sources of variation:

- **Common product demand changes:** demand changes that affect many firms producing the same product.
- **Idiosyncratic destination demand changes:** demand changes that affect a firm because it happened to sell a product to particular destinations before the recession.

For example, suppose two Portuguese firms both export textiles. If world demand for textiles falls everywhere, both firms face a common product shock. But if one firm sold textiles mainly to a country where demand collapsed and another sold to a country where demand was stable, then the first firm faces an idiosyncratic shock.

Interpretation: This decomposition is the paper's key measurement contribution. A generic export shock does not automatically identify firm-specific demand. Some of it may reflect shocks shared by close competitors.

2.10 Common demand component

The common component captures the part of the export demand shock that is shared across firms producing the same product. Let ΔM_m denote the average foreign import-demand change for product m across destination countries. Then:

$$\Delta \tau_j^{com} = \sum_{m \in M} s_{jm}^{pre} \Delta M_m.$$

This component depends on the firm’s pre-recession product mix, but not on the specific destinations to which the firm exported those products.

Interpretation: $\Delta\tau_j^{com}$ captures product-level demand changes. Because many firms may produce the same product, this component can affect not only firm j but also its competitors.

2.11 Idiosyncratic demand component

The idiosyncratic component captures the within-product, across-destination part of the shock:

$$\Delta\tau_j^{id} = \sum_{m \in M} \sum_{c \in C} s_{jmc}^{pre} (\Delta M_{mc} - \Delta M_m).$$

Equivalently:

$$\Delta\tau_j^{id} = \Delta\tau_j^{tot} - \Delta\tau_j^{com}.$$

This component asks whether firm j was exposed to destination markets that performed better or worse than the average destination market for the same product.

Interpretation: $\Delta\tau_j^{id}$ is the part of the shock closest to firm-specific demand. It varies because firms selling the same product were connected to different destination countries before the recession.

2.12 Close competitors

The paper defines close competitors using local product-market proximity. A close competitor is a firm in the same municipality and four-digit industry.

For firm j , let $g(j)$ denote its municipality-industry cell. The competitor shock is constructed from other firms in the same cell:

$$\Delta\tau_{-j} = \frac{1}{N_{g(j)} - 1} \sum_{\ell \neq j, \ell \in g(j)} \Delta\tau_\ell.$$

The paper uses these competitor shocks to test whether a shock is truly firm-specific or whether it affects a broader local market.

Interpretation: If the idiosyncratic shock is truly firm-specific, firm j ’s idiosyncratic shock should not predict outcomes at close competitors. If the common shock is market-wide, it should predict competitor outcomes.

2.13 Firm controls

The empirical design includes pre-period firm controls interacted with year indicators. These controls are measured before the shock and help absorb differential trends across firms with different baseline characteristics.

Typical controls include:

- pre-period firm size;
- pre-period sales or output;
- pre-period exports;
- export intensity;

- destination exposure;
- product composition;
- exposure to major destinations such as Spain and Angola.

Interpretation: These controls reduce concern that the shock is proxying for pre-existing differences between firms, such as size, exporter sophistication, or specialization in particular foreign markets.

2.14 Spain and Angola exposure

Spain and Angola receive special attention because they are especially important destinations for Portuguese exporters. Firms that export heavily to these destinations may differ from other exporters for reasons unrelated to recession demand shocks.

The paper therefore controls for year-specific effects of pre-period exposure to these destinations:

$$Share_{j,Spain}^{pre} \times \mathbf{1}\{t = k\},$$

$$Share_{j,Angola}^{pre} \times \mathbf{1}\{t = k\}.$$

Interpretation: These controls help ensure that the results are not simply comparing firms with different historical links to Spain or Angola.

2.15 Worker controls and fixed effects

In worker-level regressions, the paper uses worker fixed effects and year fixed effects. A stylized worker-level specification is:

$$w_{it} = \alpha_i + \alpha_t + \beta \Delta \tau_{j(i,2007)} \times Post_t + Controls_{it} + \varepsilon_{it}.$$

Here:

- α_i are worker fixed effects;
- α_t are year fixed effects;
- $\Delta \tau_{j(i,2007)}$ is the shock to the worker's 2007 employer.

Interpretation: Worker fixed effects absorb permanent worker ability, education, occupation quality, and other time-invariant worker characteristics. The wage effect is identified from changes in the same worker's wage path after the employer-specific demand shock.

2.16 Attached-incumbent sample

One important worker sample consists of attached incumbents: workers who remain at their 2007 employer. For these workers, wage changes are measured within the same employment relationship.

$$w_{it}^{attached} = \alpha_i + \alpha_t + \beta \Delta \tau_{j(i,2007)} \times Post_t + \varepsilon_{it}.$$

Interpretation: This sample is useful because it isolates wage changes for continuing workers. It is less affected by selective hiring, layoffs, or changes in workforce composition.

2.17 Employment and worker-flow measures

The paper also studies whether firm demand shocks affect employment and worker flows. Useful worker-flow outcomes include:

- whether the worker stays with the 2007 employer;
- whether the worker separates from the firm;
- whether the worker is employed at another firm;
- whether the firm hires new workers;
- whether the firm exits.

A simple employment-flow decomposition is:

$$Emp_{jt} = Incumbents_{jt} + NewHires_{jt} - Separations_{jt}.$$

Interpretation: These outcomes help distinguish wage pass-through from employment adjustment. If shocks mainly affect hiring rather than incumbent separations, wage effects for attached workers are less likely to be driven by selective retention.

2.18 Outcome transformations

Many outcomes are measured in logarithms or growth rates. For example:

$$\Delta \log Sales_{jt} = \log Sales_{jt} - \log Sales_{j,2007},$$

$$\Delta \log w_{it} = \log w_{it} - \log w_{i,2007}.$$

When the paper uses levels or ratios, the interpretation changes accordingly. Log outcomes can be read approximately as percentage changes.

Interpretation: The use of log outcomes makes coefficients easier to interpret as elasticities or semi-elasticities. It also reduces the influence of very large firms or very high-wage workers.

2.19 Why 2007 employer assignment matters

For worker outcomes, treatment is assigned based on the worker's 2007 employer:

$$Treatment_i = \Delta \tau_{j(i,2007)}.$$

This assignment is fixed before the recession. Even if a worker later moves to another firm, the worker's treatment exposure remains the shock to the original 2007 employer.

Interpretation: This avoids endogenous reassignment of treatment after the shock. If workers who leave badly shocked firms were reassigned to their new employers' shocks, job mobility itself would contaminate the treatment variable.

2.20 Measurement caveats

The paper's measures are powerful but not perfect.

- Export demand shocks are predicted shocks, not directly observed firm demand curves.

- Foreign import demand may contain global supply-chain changes, not only final demand.
- The idiosyncratic component is closer to firm-specific variation, but it may still interact with domestic firm characteristics.
- Worker wage outcomes are observed only for workers who remain in administrative employment records.
- Firm exit can mechanically remove some firm-level outcomes from the sample.
- The common component is useful, but it is harder to interpret as purely firm-specific because it can affect competitors and local labour markets.

Interpretation: These caveats explain why the paper relies on multiple validation exercises: event studies, competitor placebo tests, controls for destination exposure, and worker fixed-effect specifications.

2.21 Bottom line

The data and measurement strategy can be summarized as:

pre-recession export portfolio + foreign recession demand changes $\Rightarrow$ predicted firm demand shock.

The key refinement is:

$$\Delta\tau_j^{tot} = \Delta\tau_j^{com} + \Delta\tau_j^{id}.$$

Interpretation: The paper’s central empirical object is not simply an export shock. It is the split between a common component that affects many firms and an idiosyncratic component that moves one firm’s demand because of its pre-recession customer-destination portfolio. This split is what allows the authors to study firm-specific rent sharing separately from broader labour-market effects.

3 Models and empirical specifications

3.1 Conceptual wage equation

$$P_j = \bar{P} p_j, \quad w_{ij}^* = \bar{w}(\bar{P}) + \beta C_j(\bar{P} p_j).$$

Interpretation: The wage equals a competitive outside option plus a bargained share of the replacement-cost surplus. Idiosyncratic demand affects wages only if replacement costs are demand-sensitive.

3.2 Wage incidence elasticities

$$\Gamma^{w,p} = \frac{\partial \ln w_{ij}^*}{\partial \ln p_j} = \frac{\beta C(P_j)}{w_{ij}^*} \frac{d \ln C(P_j)}{d \ln P_j}.$$

$$\Gamma^{w,\bar{P}} = \frac{\beta C(P_j)}{w_{ij}^*} \frac{d \ln C(P_j)}{d \ln P_j} + \frac{\bar{w}(\bar{P})}{w_{ij}^*} \frac{d \ln \bar{w}(\bar{P})}{d \ln \bar{P}}.$$

Interpretation: The idiosyncratic elasticity isolates the rent-sharing channel; the common elasticity mixes rent sharing with labour-market outside-option effects.

3.3 Shock construction and pass-through

$$\Delta\tau_j = \sum_{m,c} s_{jmc}^{pre} \Delta M_{mc}^{-PT}, \quad \Delta\tau_j = \Delta\tau_j^{com} + \Delta\tau_j^{id}.$$

$$w_{it} = \alpha_i + \alpha_t + \beta \Delta\tau_{j(i)} \times Post_t + X'_{it} \gamma + \nu_{it}.$$

$$\text{Pass-through IV: } w_{it} = \alpha_i + \alpha_t + \Gamma Y_{j(i)t} \times Post_t + X'_{it} \gamma + \nu_{it},$$

where $Y_{j(i)t} \times Post_t$ is instrumented by $\Delta\tau_{j(i)} \times Post_t$.

Interpretation: The reduced form identifies wage effects of shocks; the IV converts those effects into wage responses for a given change in firm output.

4 Identification and empirical design

The paper's central empirical challenge is to distinguish two different reasons why wages may respond to a firm's product-demand shock. A firm's workers may receive higher wages because their own employer's output demand rises, which is a firm-specific rent-sharing channel. But wages may also rise because many nearby or similar employers are hit by the same product-market shock, increasing outside options and wage competition in the broader labour market. The paper's identification strategy is designed to separate these two forces.

4.1 Core identification idea

The empirical design uses the Great Recession as a source of unexpected foreign demand shocks for Portuguese exporters. Portuguese firms sold different products to different foreign destinations before the recession. When foreign import demand collapsed in 2008–09, firms were therefore exposed to different demand changes depending on their pre-recession export portfolio.

The key identifying variation is:

$$\text{pre-recession export exposure}_{jmc} \times \text{foreign import-demand change}_{mc}.$$

Here, j indexes Portuguese firms, m indexes products, and c indexes destination countries. The crucial point is that the foreign demand shocks are measured outside Portugal. The authors exclude imports from Portuguese firms when measuring destination-product import demand, so the shock is not mechanically caused by the exporting performance of the Portuguese firms in the sample.

Interpretation: The design is a shift-share design, but the identifying assumption is not that all shift-share instruments are automatically exogenous. Identification comes from the unexpected country-product import-demand shifts generated by the global recession, interacted with pre-determined export links from 2005–07.

4.2 Construction of the baseline firm-level demand shock

Let s_{jmc}^{pre} be firm j 's pre-recession share of exports going to product m in destination country c :

$$s_{jmc}^{pre} = \frac{Exports_{jmc}^{2005-2007}}{\sum_{\mu \in M} \sum_{\kappa \in C} Exports_{j\mu\kappa}^{2005-2007}}.$$

Let ΔM_{mc} be the change in non-Portuguese import demand for product m in country c between the pre-recession period and the recession trough. The paper measures this using the symmetric growth rate:

$$\Delta M_{mc} = \frac{NPI_{mc}^{post} - NPI_{mc}^{pre}}{\frac{1}{2}(NPI_{mc}^{post} + NPI_{mc}^{pre})},$$

where NPI_{mc} denotes non-Portuguese imports of product m by country c .

The baseline firm-level export demand shock is then:

$$\Delta \tau_j^{tot} = \sum_{m \in M} \sum_{c \in C} s_{jmc}^{pre} \Delta M_{mc}.$$

Interpretation: $\Delta \tau_j^{tot}$ is a predicted export-demand change. It asks: if firm j kept the same pre-recession product-destination portfolio, how much would its foreign demand change because of external import-demand shifts abroad?

4.3 Why the baseline firm-level shock is not automatically firm-specific

A key contribution of the paper is to show that a firm-level export demand shock can contain both firm-specific and market-wide components. Suppose two Portuguese firms produce the same product but sell it to different countries. If one country's import demand falls more than another's, that creates relatively idiosyncratic variation across firms. But if global demand for the product falls everywhere, all Portuguese producers of that product may be affected.

Therefore, the baseline shock is decomposed as:

$$\Delta \tau_j^{tot} = \Delta \tau_j^{com} + \Delta \tau_j^{id}.$$

The common component is based on product-level demand changes common across destinations:

$$\Delta \tau_j^{com} = \sum_{m \in M} s_{jm}^{pre} \Delta M_m,$$

where

$$s_{jm}^{pre} = \sum_{c \in C} s_{jmc}^{pre}$$

is firm j 's pre-recession export share in product m , and ΔM_m is the average global import-demand change for product m .

The idiosyncratic component is the within-product, destination-specific residual:

$$\Delta \tau_j^{id} = \sum_{m \in M} \sum_{c \in C} s_{jmc}^{pre} (\Delta M_{mc} - \Delta M_m).$$

Equivalently:

$$\Delta \tau_j^{id} = \Delta \tau_j^{tot} - \Delta \tau_j^{com}.$$

Interpretation: The common component captures demand movements that affect all producers of a product. The idiosyncratic component captures whether firm j happened to sell to countries where demand for its product fell more or less than the product's global average. This is the component closest to a firm-specific demand shock.

4.4 Difference-in-differences structure

The main firm-level specification compares firms with different pre-determined shock exposure before and after the Great Recession:

$$Y_{jt} = \alpha_t + \delta_j + \beta \Delta \tau_j \times Post_t + \sum_{k \neq 2007} \gamma_k X_j^{pre} \times \mathbf{1}\{t = k\} + \nu_{jt}.$$

Here:

- Y_{jt} is a firm-level outcome, such as exports, sales, value added, payroll, or employment;
- α_t are year fixed effects;
- δ_j are firm fixed effects;
- $\Delta \tau_j$ is one of the demand shock measures: $\Delta \tau_j^{tot}$, $\Delta \tau_j^{com}$, or $\Delta \tau_j^{id}$;
- $Post_t$ is an indicator for the post-recession years;
- X_j^{pre} are pre-period controls interacted with year indicators.

The baseline pre-period is 2006–07, the post-period is 2009–11, and 2008 is treated as a transition year.

Interpretation: The coefficient β measures how much a firm's outcome changes after the recession for a given predicted change in export demand. Firm fixed effects absorb permanent firm differences, and year fixed effects absorb aggregate Portuguese trends.

4.5 Dynamic event-study specification

To test for pre-trends and to examine timing, the paper also estimates:

$$Y_{jt} = \alpha_t + \delta_j + \sum_{k \neq 2007} \beta_k \Delta \tau_j \times \mathbf{1}\{t = k\} + \sum_{k \neq 2007} \gamma_k X_j^{pre} \times \mathbf{1}\{t = k\} + \nu_{jt}.$$

The omitted year is 2007, just before the recession. The coefficients β_k trace out the effect of the shock year by year.

Interpretation: If the demand shock is valid, the pre-2008 coefficients should be close to zero. Large pre-trends would suggest that firms exposed to worse recession shocks were already on different paths before the shock occurred.

4.6 Controls for potentially non-random export exposure

A concern is that firms selecting into certain export destinations may differ systematically from firms exporting elsewhere. For example, Portuguese firms exporting heavily to Spain or Angola may differ from other firms because Spain is geographically close and Angola is a major former colony.

The paper therefore includes controls for year-specific effects of:

pre-period export share to Spain,

pre-period export share to Angola.

In additional specifications, the authors also control for year-specific effects of pre-period export activity, including exports in logs, exports in levels, and exports as a share of sales.

Interpretation: These controls reduce the risk that the estimates are driven by persistent differences between firms that exported to particular destinations, rather than by the recession-induced demand changes in those destinations.

4.7 Sorting test using firms with multiple destinations

The paper uses a within-firm test to assess whether better firms systematically chose better performing export markets. For firms selling the same product to multiple destinations, the authors test whether export changes to each destination are predicted by non-Portuguese import growth in that same destination-product market.

A simplified version of the relationship-level test is:

$$\Delta Exports_{jmc} = \alpha_j + \beta \Delta M_{mc} + \varepsilon_{jmc}.$$

The important feature is the firm fixed effect α_j . This compares different destination markets within the same firm.

Interpretation: If unobserved firm quality were the main driver, adding firm fixed effects would greatly weaken the relationship between destination import demand and firm export growth. The paper finds that the relationship remains strong, supporting the interpretation that foreign demand movements are causal shocks rather than sorting on firm quality.

4.8 Validation of the common/idiosyncratic decomposition

The decomposition generates sharp predictions.

First, the idiosyncratic component should affect the shocked firm's exports and sales:

$$\Delta \tau_j^{id} \Rightarrow \Delta Exports_j, \Delta Sales_j.$$

Second, the idiosyncratic component should not affect close competitors:

$$\Delta \tau_j^{id} \nRightarrow \Delta Sales_{-j}.$$

Third, the common component should affect both the firm and its close competitors:

$$\Delta \tau_j^{com} \Rightarrow \Delta Sales_j, \Delta Sales_{-j}.$$

The paper defines close competitors as firms in the same municipality and four-digit industry. For each firm j , the close-competitor placebo shock is the average shock among other firms in the same municipality-industry cell:

$$\Delta \tau_{-j} = \frac{1}{N_{g(j)} - 1} \sum_{\ell \neq j, \ell \in g(j)} \Delta \tau_\ell,$$

where $g(j)$ is firm j 's local industry cell.

Interpretation: This is one of the most important validation exercises. If the idiosyncratic component is truly firm-specific, it should not move outcomes at rival employers. If the common component is a market-level shock, it should spill over to those rivals. The paper finds exactly this pattern.

4.9 Worker-level incidence design

The worker-level analysis follows the workers employed at shocked firms in 2007. This fixed cohort design is important because it separates wage changes for incumbent workers from changes in workforce composition.

A stylized worker-level specification is:

$$w_{it} = \alpha_i + \alpha_t + \beta \Delta \tau_{j(i,2007)} \times Post_t + \sum_{k \neq 2007} \gamma_k X_{j(i,2007)}^{pre} \times \mathbf{1}\{t = k\} + \varepsilon_{it},$$

where:

- w_{it} is worker i 's wage outcome;
- α_i are worker fixed effects;
- α_t are year fixed effects;
- $j(i, 2007)$ is worker i 's employer in 2007;
- $\Delta \tau_{j(i,2007)}$ is the 2007 employer's demand shock.

Interpretation: Worker fixed effects ensure that the estimates are not driven by stable worker quality differences. The design asks whether the same worker's wage path changes when the worker's pre-recession employer receives a demand shock.

4.10 Attached-incumbent wage design

A central wage outcome is the wage of attached incumbents, meaning workers who remain with their 2007 employer. This focuses on within-firm wage changes for continuing employment relationships.

$$w_{it}^{attached} = \alpha_i + \alpha_t + \beta \Delta \tau_{j(i,2007)} \times Post_t + Controls_{it} + \varepsilon_{it}.$$

Interpretation: This specification is designed to test rent sharing within ongoing employment relationships. If idiosyncratic demand shocks change wages of attached incumbents, the effect cannot be explained only by the firm hiring different types of workers after the shock.

4.11 Employment adjustment and composition

The paper separately studies employment adjustment, hiring, and separations. The purpose is to verify whether wage effects are contaminated by large changes in who remains employed.

A useful decomposition is:

$$\frac{\Delta Emp_{jt}}{Emp_j^0} = \frac{Hires_{jt}}{Emp_j^0} + \frac{Retentions_{jt}}{Emp_j^0} - 1.$$

Interpretation: The paper finds that idiosyncratic shocks mainly affect hiring, not incumbent separation. This matters because it supports the interpretation that attached-incumbent wage effects are not mostly driven by selective layoffs. In the Portuguese setting, strict dismissal protections make layoffs rare and make incumbent relationships especially important.

4.12 Pass-through IV design

The reduced-form wage regressions show whether shocks affect wages. The pass-through IV specification translates those shock effects into the effect of firm output or sales on wages.

The first stage is:

$$Y_{jt} = \alpha_j + \alpha_t + \pi \Delta \tau_j \times Post_t + X'_{jt} \gamma + u_{jt},$$

where Y_{jt} is firm output or sales.

The second stage is:

$$w_{it} = \alpha_i + \alpha_t + \Gamma \widehat{Y}_{j(i,2007)t} + X'_{it} \delta + \varepsilon_{it}.$$

Equivalently, the paper instruments firm output changes with export-demand shocks:

$$Y_{j(i,2007)t} \times Post_t \text{ instrumented by } \Delta \tau_{j(i,2007)} \times Post_t.$$

Interpretation: The pass-through coefficient Γ is the wage response to a firm-output change induced by an external demand shock. This is more interpretable than the reduced form because it converts a demand-shock coefficient into an elasticity of wages with respect to firm performance.

4.13 Why idiosyncratic shocks identify internal rent sharing

The idiosyncratic component is central because it affects one firm's product demand without systematically changing the product-market or labour-market conditions faced by nearby rivals.

If wages respond to $\Delta \tau_j^{id}$, the interpretation is:

firm-specific demand shock $\Rightarrow$ own-firm output change $\Rightarrow$ own workers' wage change.

This is evidence of internal rent sharing or bargaining over firm-specific quasi-rents.

Interpretation: In a perfectly competitive labour market, a firm-specific demand shock should not change wages if workers can freely move and firms take wages as given. A wage response to idiosyncratic demand therefore indicates that firm-specific conditions matter for wage setting.

4.14 Why common shocks identify labour-market competition effects

The common component affects many producers in the same product market. It may also affect competitors in the same local labour market.

If wages respond to $\Delta \tau_j^{com}$, the interpretation is:

market-wide demand shock $\Rightarrow$ many employers affected $\Rightarrow$ outside options and wage competition change.

Interpretation: The common component mixes firm demand with broader labour-market pressure. This is why the paper separates it from the idiosyncratic component. Without the decomposition, a generic firm-level export shock could be mistaken for firm-specific rent sharing even when it partly reflects market-wide wage competition.

4.15 Clustering and inference

The level of clustering depends on the shock component.

- For idiosyncratic shocks, standard errors are clustered at the firm level because the identifying variation is firm-specific.
- For common shocks, standard errors are clustered at the four-digit industry level because the common component varies at a broader product-market level.
- In relationship-level validation tests, standard errors are clustered in ways that account for correlation across firm-market observations.

Interpretation: Correct clustering is important because treatment is not randomly assigned at the individual worker level. Workers inside the same firm share the same demand shock, and firms producing similar products can share the same common shock.

4.16 Key identifying assumptions

The design relies on several assumptions.

- **Shock relevance.** Pre-recession export exposure interacted with foreign import-demand changes must predict actual firm exports and sales.
- **Parallel trends.** Absent the Great Recession demand shifts, firms with different shock exposure would have followed similar outcome trends.
- **No endogenous destination sorting on future shocks.** Firms did not choose their 2005–07 export destinations based on information about the 2008–09 recession demand collapse.
- **Idiosyncratic component is firm-specific.** The idiosyncratic component should not systematically affect close competitors in the same municipality and industry.
- **Common component is market-wide.** The common component can affect rival firms and therefore should not be interpreted as purely internal rent sharing.
- **Stable worker composition for wage analysis.** Worker fixed effects and the attached incumbent sample must absorb or avoid composition changes that would otherwise contaminate wage effects.

Interpretation: The paper does not assume that all export demand shocks are firm-specific. The key innovation is to test which part is firm-specific and which part is common across product or labour markets.

4.17 Threats to identification and how the paper addresses them

- **Sorting into destinations.** Firms that exported to badly hit destinations may have been different before the recession. The paper addresses this with within-firm destination tests, controls for pre-period export patterns, and event-study pre-trend checks.
- **Spain and Angola exposure.** Spain and Angola are special export destinations for Portuguese firms. The paper controls for year-specific effects of pre-period exposure to both destinations.
- **Product-market confounds.** Some product demand shocks affect all firms in a product market. The paper separates these common shocks from destination-specific idiosyncratic shocks.
- **Labour-market spillovers.** Idiosyncratic shocks could in principle affect other employers through local wage pressure. The close-competitor tests show that the idiosyncratic component does not predict competitor outcomes, while the common component does.
- **Firm exit and worker attrition.** Negative common shocks can cause firm exit and worker separation. The paper studies both the full sample and samples of never-exiter firms, and it distinguishes attached-incumbent wage changes from employment composition effects.
- **Domestic demand.** Export shocks could be correlated with domestic demand for the same products. The idiosyncratic component is less vulnerable to this concern because it is based on within-product differences across destination countries.

Interpretation: The identification strategy is strongest for the idiosyncratic component. The common component is economically important but harder to interpret as purely causal firm-specific variation because it can move product markets and labour markets more broadly.

4.18 How to read the main estimates

For firm outcomes, a coefficient on $\Delta\tau_j \times Post_t$ measures how much firm sales, exports, or output changes after the recession for a one-unit predicted demand shock. For wage outcomes, the coefficient measures how much wage growth changes for workers whose 2007 employer was exposed to the shock.

The main interpretation is:

$$\Delta\tau_j^{id} \rightarrow \Delta Sales_j \rightarrow \Delta Wages_{ij} \quad \text{but not} \quad \Delta Wages_{-j}.$$

This pattern supports the claim that firm-specific demand rents pass through to workers.

By contrast:

$$\Delta\tau_j^{com} \rightarrow \Delta Sales_j \quad \text{and} \quad \Delta Sales_{-j},$$

so common shocks are interpreted as broader product-market or labour-market shocks.

Interpretation: The empirical design is built around this contrast. Idiosyncratic shocks identify internal firm wage-setting responses; common shocks reveal how broader market conditions and outside options affect wages.

4.19 Bottom line

The design combines three layers of evidence.

1. **External demand shocks:** foreign import-demand changes during the Great Recession generate plausibly exogenous variation in firm demand.
2. **Common/idiosyncratic decomposition:** the paper separates shocks shared by product markets from shocks specific to firms' pre-recession destination choices.
3. **Validation using close competitors and event studies:** idiosyncratic shocks affect own firms but not close competitors, while common shocks affect both; event studies show limited evidence of pre-trends.

Interpretation: This is why the wage results can be interpreted as evidence that firm-specific demand shocks pass through to incumbent workers' wages. The most important empirical object is not a generic export shock, but the idiosyncratic component of export demand that moves one firm's output without moving nearby competitors.

5 Tables and figures

5.1 Figure 1: Growth and exports in Portugal around the Great Recession; Table 1: Summary statistics: pre-period exports of sample firms

Read: Figure 1 motivates the recession timing; Table 1 shows the analysis sample has meaningful export exposure across destinations and products.

Detailed interpretation: The figure shows the external trade collapse that supplies the demand shock. Table 1 shows why firm-level shock construction is feasible: firms sell to multiple destination-product markets.

How it fits the paper: These visuals anchor the setting and the sample.

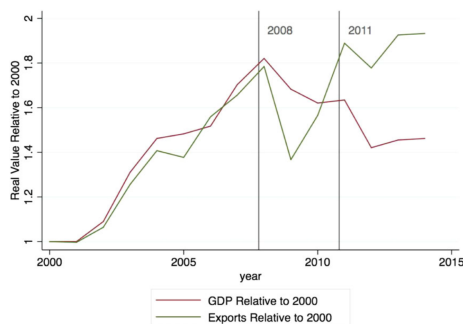


FIGURE 1
Growth and exports in Portugal around the Great Recession

Source: The World Bank (2016). Figure plots annual GDP and total exports for Portugal in real euros, with each variable indexed to its 2000 level. Vertical lines indicate the start of the Great Recession in the U.S. at the end of 2007 and the beginning of the Portuguese sovereign debt crisis in the spring of 2011.

(a) Figure 1: Growth and exports in Portugal around the Great Recession. The figure highlights the externally driven export contraction before Portugal's later sovereign-debt crisis.

TABLE 1
Summary statistics: pre-period exports of sample firms

	Analysis sample				Large exporters		
	Mean	P25	P50	P75	SD	Mean	P50
2007 exports, euros	1,132,340	41,718	321,375	1,187,745	2,364,478	18,500,000	5,312,642
Pre-period export exposure	0.342	0.027	0.201	0.650	0.345	0.464	0.464
# Destination countries	5.21	2	3	6	5.7	14.27	10
# Major (2D) products	3.6	1	2	4	4.9	6.9	4
# Detailed (6D) products	10.3	2	4	10	22.0	25.1	11
Number of firms	4173					934	

Notes: Table displays export statistics for firms appearing in the export data in each of 2005, 2006, and 2007, tabulated from the firm-product-destination-year level data. Analysis sample contains all such firms with 100 or fewer employees during pre-period (2005–7 average). Large exporters are remainder of firms with over 100 employees. Exports are measured in constant 2007 euros. Exports/sales is the ratio of total exports to total sales from the balance sheet data (a distinct source) averaged across years 2005–7. Counts of destination countries and products (HS2 and HS6) pool 2005, 2006, and 2007 exports of each firms to reflect construction of the shock.

(b) Table 1: Summary statistics: pre-period exports of sample firms. The table documents export scale, destinations, and product diversification used to build firm shocks.

5.2 Table 2: Comparison of firms and workers in sample and population; Table 3: Test of sorting for firms with multiple destinations

Read: Table 2 compares sample firms and workers to population data. Table 3 shows foreign import demand predicts firm-market export changes.

Detailed interpretation: Table 2 is a representativeness check. Table 3 is a first-stage and sorting validation: pre-existing firm-market exposure responds to external demand movements.

How it fits the paper: These tables support the sample construction and the relevance of the demand shifter.

TABLE 2
Comparison of firms and workers in sample and population

	Analysis sample					Full count data				
	Mean	P10	P50	P90	SD	Mean	P10	P50	P90	SD
<i>Firms</i>										
Employees	27.9	5	21	64	23.16	3.75	0	0	6	51.67
Sales/worker if Emp > 0, euros	173,226	33,803	103,085	374,312	238,679	98,055	0	29,892	180,425	2,894,556
Value added/worker	34,769	11,935	26,575	61,969	40,939	20,397	0	11,019	42,658	203,807
if Emp > 0, euros										
N firms, Emp > 0	4173				278,226					
N firms					714,212					
<i>Workers</i>										
Monthly wage, euros	760.80	403	575	1310	533.58	753.12	403	561.74	1318	530.46
Hourly wage, euros	4.45	2.34	3.35	7.67	3.14	4.52	2.33	3.33	8.15	3.27
Log monthly wage	6.49	6.00	6.35	7.18	0.49	6.47	6	6.33	7.18	0.50
Log hourly wage	1.34	0.85	1.21	2.04	0.49	1.35	0.85	1.20	2.10	0.50
Fixed term, percent of sample	0.20				0.25					
Tenure, months (all workers)	121	9	93	260	105.82	90.01	4.00	57	227	98.58
Female, percent of sample	0.44				0.43					
Regular hours per month	171	162	173	176	7.55	154	173	176		11
N workers		115,526			2,490,452					

Notes: Table compares firms and workers in the analysis sample to the population data for year 2007. All figures in currency units are in 2007 euros. Data on firm sales and output are from the balance sheet data (IES) with $N = 278,226$. Employment counts are tabulated from the matched employer-employee data (QP) for firms that appear in both the QP and IES, $N = 278,226$. Worker statistics are tabulations of employee records in the cleaned QP, restricting to the highest paying full-time job (more than 120 hours per month) per worker, $N = 2,490,452$. "Monthly Wage" and "Hourly Wage" are the base contract pay earned during regular hours during the reference month in the QP, excluding overtime, fringe payments, and bonuses. "Percent FTC" is the percent of workers at the firm with fixed-term contracts.

(a) Table 2: Comparison of firms and workers in sample and population.

TABLE 3
Test of sorting for firms with multiple destinations

	Outcome is change in exports by firm j of product m to country c , measured by				
	Symmetric growth rate	Log change (if > 0)	Any exports		
	(1)	(2)	(3)	(4)	(5)
Coefficient: change in destination c imports of product module m	0.0600*** (0.0152)	0.0677*** (0.0112)	0.476*** (0.0481)	0.397*** (0.0507)	0.0179 (0.0154)
Baseline controls	x	x	x	x	x
Firm fixed effects	x	x	x	x	x
N	101,344	101,344	35,193	34,804	101,344

Notes: Table reports results from regressions of changes in exports by firm j of product m to destination c on the change (symmetric growth rate) of imports of m to country c from all other countries (excluding Portugal) during the same period, as in [Supplementary Material, Equation \(A.11\)](#). Observations are firm-market pairs (markets are country $\times$ six-digit product). Sample includes all firms in primary analysis sample with exports to at least two distinct markets in the pre-period. Changes are taken from 2006–7 to 2009–10. When no exports occur in the post-period, log values are treated as missing. Regressions are unweighted. Standard errors are two-way clustered at the firm and market level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

(b) Table 3: Test of sorting for firms with multiple destinations.

5.3 Figure 2: Year-specific effects on exports and sales in common currency units; Table 4: Effects on firm sales and output

Read: Both visuals show that shocks move exports and sales after 2007, with limited pre-trends.

Detailed interpretation: The idiosyncratic component affects own-firm sales; the common component has broader and more persistent effects.

How it fits the paper: These visuals establish the first stage for wage pass-through.

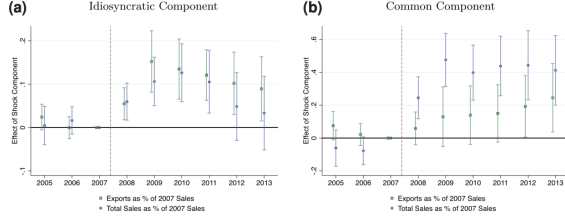


FIGURE 2
Year-specific effects on exports and sales in comparable currency units
Notes: Figure displays year-specific effects of the specified shock to export demand Δ_j^f on sales and exports in common euro units, normalised by firms' 2007 total sales. Sample is full analysis sample of firms ($N = 4173$). Year-specific coefficients and 95% confidence intervals are from regressions on interactions of Δ_j^f and an indicator for each year, with all interactions estimated jointly as in Equation (13). Estimates for each outcome are from separate regressions. Confidence intervals are based on standard errors that are clustered at the firm level when Δ_j^f is the dependent variable and at the four-digit industry level when Δ_j^{com} is the dependent variable to account for potential serial correlation of errors. Exports are coded as zero in years when firms do not appear in the export data. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the Δ_j^f is the shock, we include a control for Δ_j^{com} . Corresponding estimates for sample of never-exit firms are presented in [Supplementary Material, Appendix Figure E.1](#).

(a) Figure 2: Dynamic effects on exports and sales.

TABLE 4
Effects on firm sales and output

	Log exports (1)	Log sales (2)	Exports/2007 sales (3)	Sales/2007 sales (4)	New exp dest (5)	New exp product (6)	Firm active (7)
<i>A. Full sample (full sample of 4173 firms)</i>							
Baseline	0.635*** (0.122)	0.234*** (0.050)	0.126*** (0.040)	0.160*** (0.037)	−0.036 (0.037)	0.049 (0.046)	0.010 (0.022)
Idiosyncratic	0.592*** (0.115)	0.143*** (0.048)	0.136*** (0.029)	0.104*** (0.033)	−0.077*** (0.034)	0.085*** (0.029)	−0.003 (0.023)
Common	1.167*** (0.221)	0.693*** (0.111)	0.129 (0.090)	0.476*** (0.083)	0.080 (0.077)	−0.114 (0.153)	0.119*** (0.055)
N firm-year obs	17,408	19,729	20,865	20,865	12,685	12,685	20,865
<i>B. Never-exiters (balanced sample of 2926 firms)</i>							
Baseline	0.669*** (0.141)	0.241*** (0.038)	0.120*** (0.037)	0.186*** (0.030)	−0.015 (0.042)	0.043 (0.046)	—
Idiosyncratic	0.624*** (0.127)	0.184*** (0.038)	0.122*** (0.030)	0.145*** (0.030)	−0.061 (0.038)	0.083*** (0.032)	—
Common	1.185*** (0.213)	0.526*** (0.102)	0.157* (0.084)	0.400*** (0.078)	0.108 (0.083)	−0.130 (0.151)	—
N firm-year obs	13,436	14,612	14,630	14,630	10,331	10,331	

Notes: Sample is either full analysis sample of firms ($N = 4173$) or sample of firms that always report positive employment ("never-exiters" $N = 2926$), as specified. Each point estimate is obtained from a separate regression. Estimates are coefficients on the interaction between the specified shock to export demand Δ_j^f and $Post_t$. "Pre" years are 2006, 2007 (pre-period) and "Post" years 2009, 2010, 2011 (post-period), 2008 is omitted. Firm-year observations with zeros are treated as missing when the outcome is in logs—therefore, the baseline sample is not a balanced panel, but the never-exiters sample is. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the Δ_j^f is the shock, we include a control for Δ_j^{com} . Standard errors are clustered at the firm level when Δ_j^f is the dependent variable and at the four-digit industry level when Δ_j^{com} or Δ_j^{tot} is the dependent variable. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

(b) Table 4: Effects on firm sales and output.

5.4 Table 5: Own-firm and close-competitor effects of shock components, full sample

Read: Idiosyncratic shocks affect own firms but not close competitors; common shocks affect both.

Detailed interpretation: This is the key validation table for the common/idiosyncratic decomposition.

How it fits the paper: The table lets wage effects of idiosyncratic shocks be interpreted as internal firm rent sharing.

TABLE 5
Own-firm and close-competitor effects of shock components, full sample

	Log sales (1)	Log value added (2)	Log payroll (3)	Log employees (4)
Mean pre-post change	-0.252	-0.194	-0.021	-0.073
A. Baseline				
Baseline	0.234*** (0.050)	0.194*** (0.045)	0.092*** (0.030)	0.075*** (0.029)
<i>N</i> firm-year obs	19,729	19,217	19,111	19,111
Placebo	0.223** (0.096)	0.148* (0.085)	0.040 (0.063)	0.076 (0.060)
<i>N</i> firm-year obs	14,857	14,490	14,408	14,408
B. Idiosyncratic component				
Idiosyncratic	0.143*** (0.048)	0.112** (0.047)	0.051 (0.033)	0.035 (0.030)
<i>N</i> firm-year obs	19,729	19,217	19,111	19,111
Placebo	0.039 (0.076)	-0.000 (0.077)	-0.022 (0.058)	0.031 (0.052)
<i>N</i> firm-year obs	14,857	14,490	14,408	14,408
C. Common component				
Common	0.693*** (0.111)	0.584*** (0.124)	0.277*** (0.085)	0.287*** (0.084)
<i>N</i> firm-year obs	19,729	19,217	19,111	19,111
Placebo	0.896*** (0.173)	0.677*** (0.183)	0.281** (0.139)	0.257** (0.130)
<i>N</i> firm-year obs	14,857	14,490	14,408	14,408

Notes: Sample is full analysis sample of firms ($N = 4173$). Each point estimate is obtained from a separate regression. The estimates in the first row of each panel are coefficients on the interaction between the specified shock and export demand Δ_j^x and $Post_t$. The close-competitor "placebo" estimates in each panel replace Δ_j^x for each firm j with the average Δ_{-j}^x taken over all other firms in the same four-digit industry and the same municipality as j , exclusive of j . "Pre" years are 2006, 2007 (pre-period) and "Post" years 2009, 2010, 2011 (post-period), 2008 is omitted. Firm-year observations with zeros are treated as missing when the outcome is in logs—therefore, the baseline sample is not a balanced panel, but the never-exiter sample is. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the Δ_j^{id} is the shock, we include a control for Δ_j^{com} . Table also presents the average (real) change in the dependent variable from pre-to-post. Standard errors are clustered at the firm level when Δ_j^{id} is the dependent variable and at the four-digit industry level when Δ_j^{com} or Δ_j^{ot} is the dependent variable. *** $p < .01$, ** $p < .05$, * $p < .10$. Corresponding estimates for sample of never-exiter firms are presented in [Supplementary Material, Appendix Table E.8](#).

Table 5: Own-firm and close-competitor effects of shock components, full sample.

5.5 Figure 3: Employment adjustment; Figure 4: Log contract wage of attached incumbents

Read: Figure 3 separates hiring from incumbent retention. Figure 4 shows incumbent wage responses.

Detailed interpretation: Wage changes are not just compositional changes from separations; attached workers' wages respond to own-firm demand.

How it fits the paper: These event studies directly support the rent-sharing interpretation.

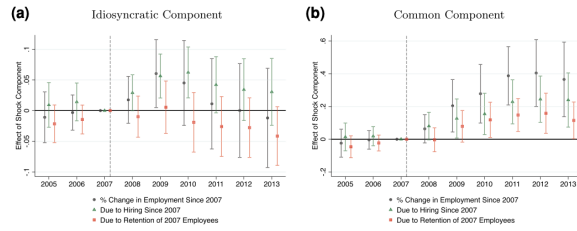


FIGURE 3

Year-specific effects on employment adjustment: hiring versus incumbent retention

Notes: Figure displays year-specific effects of the specified shock to export demand Δ_j^x on employment, total accumulated hires, and total retentions, all as a percent of firms' 2007 total employment. Sample is full analysis sample of firms ($N = 4173$). Year-specific coefficients and 95% confidence intervals are from regressions on interactions of Δ_j^x and an indicator for each year, with all interactions estimated jointly as in Equation (13). Estimates for each outcome are from separate regressions. Confidence intervals are based on standard errors that are clustered at the firm level when Δ_j^{id} is the dependent variable and at the four-digit industry level when Δ_j^{com} is the dependent variable to account for potential serial correlation of errors. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the Δ_j^{id} is the shock, we include a control for Δ_j^{com} . The change "Due to Hiring" is the total number of workers at the firm who were not present in 2007, as a percentage of 2007 employment; change "Due to Retention" is the number of employees at the firm who were employed at the firm in 2007, also as a percentage of 2007 employment. By construction, changes in these two variables related to 2007 sum to the total percent change in employment since 2007. Corresponding estimates for sample of never-exiter firms are presented in [Supplementary Material, Appendix Figure E.2](#).

(a) Figure 3: Employment adjustment, hiring versus incumbent retention.



FIGURE 4

Dynamic effects on log contract wage of attached incumbents

Notes: Sample is all individuals i whose primary full-time job (120–200 hours per month) in 2007 was at a firm j in the main analysis sample of 4173 firms. Figure shows year-specific reduced-form effects of the specified shock to export demand Δ_j^x on individual hourly wages conditional on remaining at the 2007 firm. Estimates for each outcome are from separate regressions. The "Other Sample Firm" estimates in each panel replace Δ_j^x for each worker i at firm $j(i)$ with the average $\Delta_{-j(i)}^x$ taken over all other close-competitor firms in the same four-digit industry and the same municipality as $j(i)$, exclusive of $j(i)$. Confidence intervals are based on standard errors that are clustered at the firm level when Δ_j^{id} is the dependent variable and at the four-digit industry level when Δ_j^{com} is the dependent variable to account for potential serial correlation of errors. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the Δ_j^{id} is the shock, we include a control for Δ_j^{com} . Corresponding estimates for 2007 employees of firms in the never-exiter sample are presented in [Supplementary Material, Appendix Figure E.3](#).

(b) Figure 4: Dynamic effects on log contract wage of attached incumbents.

5.6 Table 6: Reduced-form effects on employee wages; Table 7: Pass-through elasticity

Read: Table 6 shows reduced-form wage effects; Table 7 converts them into output pass-through elasticities around 0.10–0.15 for idiosyncratic shocks.

Detailed interpretation: OLS elasticities are much smaller, indicating attenuation in non-experimental output-wage correlations.

How it fits the paper: These tables quantify the main result.

TABLE 6
Reduced-form effects on employee and close-competitor employee wages, full sample

	Average log wage, within-spell		Average log wage, including moves		Monthly wage, including zeros		
	Monthly (1)	Hourly (2)	Monthly (3)	Hourly (4)	IHS (5)	Arc Δ (6)	At firm (7) Any job (8)
<i>A. Idiosyncratic component</i>							
Idiosyncratic	0.022*** (0.008)	0.019** (0.008)	0.027*** (0.008)	0.024*** (0.008)	0.041 (0.050)	0.033 (0.037)	−0.003 (0.025) 0.004 (0.018)
<i>N</i>	432,845	432,845	485,055	485,043	582,975	582,975	582,975
Placebo	0.010 (0.012)	0.002 (0.012)	0.013 (0.012)	0.005 (0.012)	−0.013 (0.087)	0.031 (0.064)	0.031 (0.041) 0.011 (0.030)
<i>N</i>	329,117	329,117	367,710	367,704	441,295	441,295	441,295
<i>B. Common component</i>							
Common	0.026* (0.014)	0.020 (0.014)	0.023 (0.016)	0.018 (0.016)	0.318*** (0.104)	0.193** (0.080)	0.181*** (0.060) 0.087** (0.036)
<i>N</i>	432,845	432,845	485,055	485,043	582,975	582,975	582,975
Placebo	0.074*** (0.026)	0.072*** (0.025)	0.074*** (0.026)	0.070*** (0.026)	0.497*** (0.153)	0.309*** (0.116)	0.202** (0.079) 0.117** (0.056)
<i>N</i>	329,117	329,117	367,710	367,704	441,295	441,295	441,295
Mean pre-post change	0.061	0.065	0.060	0.064	−0.487	−0.342	−0.277 −0.178

Notes: Sample is all individuals i whose primary full-time job (120–200 hours per month) in 2007 was at a firm j in the main analysis sample of 4173 firms. Each point estimate is obtained from a separate regression. The estimates in the first row of each panel are coefficients on the interaction between the specified shock to export demand $\Delta_{j(i)}^x$ and $Post_t$. The close-competitor “placebo” estimates in each panel replace $\Delta_{j(i)}^x$ for each worker i at firm $j(i)$ with the average $\Delta_{-j(i)}^x$ taken over all other firms in the same four-digit industry and the same municipality as $j(i)$, exclusive of $j(i)$. “Pre” years are 2006, 2007 (pre-period) and “Post” years 2009, 2010, 2011 (post-period), 2008 is omitted. Worker-year observations with zeros are treated as missing when the outcome is in logs, and observations when workers are not employed by their 2007 employer are treated as missing in “Within-Spell” specifications. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the $\Delta_{j(i)}^d$ is the shock, we include a control for $\Delta_{j(i)}^{com}$. Table also presents the average (real) change in the dependent variable from pre-to-post. Standard errors are clustered at the firm level when $\Delta_{j(i)}^d$ is the dependent variable and at the four-digit industry level when $\Delta_{j(i)}^{com}$ is the dependent variable. IHS denotes inverse hyperbolic sine. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Corresponding estimates for sample of 2007 employees of never-exiter firms are presented in [Supplementary Material, Appendix Table E.9](#).

(a) Table 6: Reduced-form effects on employee and close-competitor wages.

TABLE 7
Pass-through elasticity: effect in wages for given change in output, full sample

	Log wage within-spell		Monthly wage, including zeros		
	Monthly (1)	Hourly (2)	IHS (3)	Arc Δ (4)	Any job (5)
<i>A. Idiosyncratic component</i>					
IV	0.134** (0.054)	0.115** (0.052)	0.245 (0.320)	0.225 (0.229)	0.026 (0.116)
<i>N</i>	430,703	430,703	552,106	552,106	552,106
First-stage F	15.05	15.05	4.51	4.51	4.51
<i>B. Common component</i>					
IV	0.050** (0.022)	0.040* (0.023)	0.285*** (0.098)	0.159** (0.070)	0.067** (0.033)
<i>N</i>	430,703	430,703	552,106	552,106	552,106
First-stage F	23.53	23.53	33.60	33.60	33.60
<i>C. OLS</i>					
OLS	0.026*** (0.003)	0.023*** (0.003)	0.213*** (0.016)	0.159*** (0.012)	0.073*** (0.005)
<i>N</i>	430,703	430,703	552,106	552,106	552,106

Notes: Table displays estimates of pass-through elasticities, obtained from IV and OLS difference-in-differences regressions of workers i ’s log wages on the sales of their 2007 employer $j(i)$. Elasticities are estimated from Equation (15) by OLS or IVs (two-stage least squares) estimation using the specified shock component $\Delta_{j(i)}^x$ interacted with $Post_t$ as an instrument for the firm output measure $Y_{j(i)}$ interacted with $Post_t$. Sample is all individuals whose primary full-time job (120–200 hours per month) in 2007 was at a firm in the main analysis sample of 4173 firms. Each point estimate is obtained from a separate regression. “Pre” years are 2006, 2007 (pre-period) and “Post” years 2009, 2010, 2011 (post-period). Worker-year observations with zeros are treated as missing when the outcome is in logs, and observations when workers are not employed by their 2007 employer are treated as missing in “Within-Spell” specifications. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the $\Delta_{j(i)}^d$ is the shock, we include a control for $\Delta_{j(i)}^{com}$. Table also presents the average (real) change in the dependent variable from pre-to-post. Standard errors are clustered at the firm level when $\Delta_{j(i)}^d$ is the dependent variable and at the four-digit industry level when $\Delta_{j(i)}^{com}$ is the dependent variable. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Corresponding estimates for sample of 2007 employees of never-exiter firms are presented in [Supplementary Material, Appendix Table E.10](#).

(b) Table 7: Pass-through elasticity: effect in wages for a given change in output.

5.7 Table 8: Robustness of idiosyncratic shock pass-through estimates

Read: The idiosyncratic pass-through estimate remains positive across alternative controls, shock definitions, and samples.

Detailed interpretation: Robustness checks reduce concern that the main result depends on one decomposition or control set.

How it fits the paper: The table strengthens the causal interpretation of firm-specific wage pass-through.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>A. Full sample</i>										
Sales elasticity IV	0.134** (0.054)	0.094** (0.048)	0.148** (0.066)	0.089** (0.041)	0.138*** (0.054)	0.127** (0.051)	0.145** (0.063)	0.123** (0.050)	0.112 (0.089)	0.151** (0.059)
Reduced-form effect	0.022*** (0.008)	0.013*** (0.006)	0.021** (0.008)	0.014** (0.006)	0.023*** (0.008)	0.021*** (0.008)	0.021** (0.008)	0.021*** (0.008)	0.010 (0.007)	0.032** (0.015)
<i>N firms</i>	4156	4156	4156	4156	4156	4156	4155	4156	4152	5204
<i>N workers</i>	109,413	109,413	109,413	109,413	109,413	109,413	109,397	109,413	109,409	389,897
<i>B. Never-exiters</i>										
Sales elasticity IV	0.102** (0.049)	0.070 (0.044)	0.102* (0.055)	0.072* (0.040)	0.103** (0.048)	0.096** (0.046)	0.101* (0.057)	0.094** (0.046)	0.070 (0.060)	0.127** (0.057)
Reduced-form effect	0.020** (0.010)	0.011 (0.007)	0.018* (0.010)	0.012* (0.007)	0.020** (0.009)	0.019** (0.009)	0.017* (0.010)	0.019** (0.009)	0.009 (0.008)	0.030* (0.017)
<i>N firms</i>	2921	2921	2921	2921	2921	2920	2921	2919	2919	3712
<i>N workers</i>	80,639	80,639	80,639	80,639	80,639	80,623	80,639	80,637	80,637	328,108
Angola/Spain exposure	x		x		x	x	x	x	x	x
Common component	x				x	x	x			
Pre-period exports				x						
Pre-period firm performance					x					
Destinations + 03-06 demand						x				
Worker attributes								x		
Main four-digit product FE									x	
Including large firms										x

Notes: Table displays robustness of IVs estimates of the pass-through of the idiosyncratic shock $\Delta_{j(i)}^d$ to wages presented in Table 7 to alternative specifications. The outcome is log monthly contract wage within-spell. Worker-year observations with zeros are treated as missing when the outcome is in logs, and observations when workers are not employed by their 2007 employer are treated as missing in "Within-Spell" specifications. Each point estimate is obtained from a separate regression. Sample in panel A is all individuals whose primary full-time job (120-200 hours per month) in 2007 was at a firm in the main analysis sample of 4173 firms; and sample in panel B is limited to those employed by the 2926 firms that never exit the sample. Column 1 replicates IV estimates from column 1 of panel A in Table 7 and includes controls for the common shock and pre-period exports to Spain and Angola—all specifications except columns 2-4 include these controls. Column 2 includes no controls, while columns 3 and 4 include the Angola/Spain controls and common shock control individually. Column 5 includes controls for firms 2005-7 exports in logs and as a share of sales. Column 6 includes controls for year-specific effects of 2005-7 average employment, sales, assets, hiring, labour productivity, wage levels, and fixed-term contract employment. Column 7 includes controls for the share of pre-period exports going to each of ten top destination countries, as well as predicted demand using 2003-6 changes in imports at baseline destinations. Column 8 includes controls for workers' tenure, age, pre-period pay, gender, and educational attainment. Column 9 includes product-class-by-year fixed effects (FE) for the four-digit HS product category that constitutes the largest share of each firm's 2005-7 exports. Column 10 replicates the baseline specification in column 1 in the extended sample including large firms. All controls are interacted with year dummies to allow for year-specific effects. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8: Robustness of idiosyncratic shock pass-through estimates.

5.8 Figure 5: Incidence of shocks by industry relationship durability; Table 9: Heterogeneity by industry relationship durability

Read: Idiosyncratic shocks matter most in low-turnover/high-durability sectors; common shocks matter more in fluid sectors.

Detailed interpretation: This is mechanism evidence: firm-specific pass-through appears where replacement costs and hold-up power are high.

How it fits the paper: These visuals tie the estimates to the theoretical turnover-cost channel.

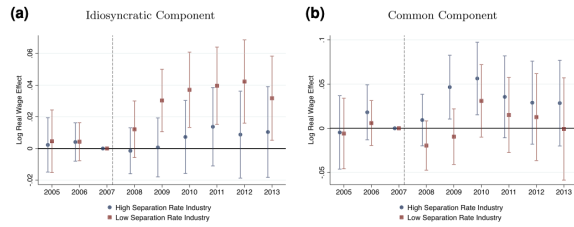


FIGURE 5
Incidence of shocks by industry relationship durability

Notes: Sample is all individuals whose primary full-time job (120-200 hours per month) in 2007 was at a firm in the main analysis sample of 4173 firms. Figure shows year-specific reduced-form effects of the interaction of the specified shock to export demand $\Delta_{j(i)}^d$ and a dummy indicating either a high separation rate industry or a low quit-rate industry on individual hourly wages conditional on remaining at the 2007 firm. Specifications included controls for year-specific effects of separation rate indicators. High (low) separation rate indicates that the firms' five-digit industry has a leave-one-out average annual separation rate of permanent contract workers (averaged across years) that is above (below) that of the median firm in the sample. All coefficients are estimated jointly using specification in Equation (16). All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005-7. Confidence intervals are based on standard errors that are clustered at the firm level when $\Delta_{j(i)}^d$ is the dependent variable and at the four-digit industry level when $\Delta_{j(i)}^{com}$ is the dependent variable to account for potential serial correlation of errors. When the $\Delta_{j(i)}^d$ is the shock, we include a control for $\Delta_{j(i)}^{com}$. Corresponding estimates for 2007 employees of firms in the never-exit sample are presented in Supplementary Material, Appendix Figure E.4.

(a) Figure 5: Incidence of shocks by industry relationship durability.

	Separation rate in industry		Typical tenure in industry		Pre-recession AKM pay-premium	
	Log hourly wage (1)	Log monthly salary (2)	Log hourly wage (3)	Log monthly salary (4)	Log hourly wage (5)	Log monthly salary (6)
<i>A. Idiosyncratic component</i>						
High frictions	0.294** (0.148)	0.310** (0.151)	0.120** (0.060)	0.146** (0.063)	0.204** (0.083)	0.225*** (0.084)
Low frictions	-0.008 (0.057)	0.011 (0.054)	0.075 (0.103)	0.079 (0.102)	0.074 (0.059)	0.087 (0.061)
Coeffs equal, p-value	0.068	0.074	0.701	0.567	0.177	0.160
<i>B. Common component</i>						
High frictions	0.010 (0.029)	0.031 (0.026)	0.028 (0.031)	0.037 (0.028)	0.019 (0.048)	0.040 (0.043)
Low frictions	0.079** (0.035)	0.079** (0.034)	0.051 (0.037)	0.067* (0.035)	0.116 (0.103)	0.072 (0.091)
Coeffs equal, p-value	0.135	0.262	0.650	0.514	0.440	0.772

Notes: Table displays estimates of subsample-specific pass-through elasticities, obtained from the interacted IV difference-in-differences regression specification in (17), where interactions of sales $Y_{j(i)}$, $Post_t$, and each of the group indicators $H_{j(i)}$ and $L_{j(i)}$ are jointly instrumented with interactions of $Post_t$, the specified shock component $\Delta_{j(i)}^d$, and each of $H_{j(i)}$ and $L_{j(i)}$. Wage outcomes are defined within-spell, and observations when workers are not employed by their 2007 employer are treated as missing. The sample is split into high and low friction subsamples based on whether each of three different measures of potential frictions is above or below median for the sample. The first measure is the leave-one-out five-digit industry average of firms' median tenure of permanent contract workers in 2003-7 (low turnover rate indicates high frictions). The second is the leave-one out average annual separation rate of permanent contract workers (averaged across years). The third is a firm-level AKM firm effect identified off of permanent contract workers who switch firms, see footnote S2 for details. All interacted specifications include controls for the categorical indicators $H_{j(i)}$ and $L_{j(i)}$ times $Post_t$. See notes to Table 7 for addition specification details. Standard errors are clustered at the firm level when $\Delta_{j(i)}^d$ is the dependent variable and at the four-digit industry level when $\Delta_{j(i)}^{com}$ is the dependent variable. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

(b) Table 9: Heterogeneity by industry relationship durability, full sample.

5.9 Table 10: Idiosyncratic shock pass-through among subgroups of workers, full sample

Read: Less-educated and blue-collar workers show larger pass-through than high-education and white-collar workers.

Detailed interpretation: The pattern suggests export demand shocks raise wages most for workers directly tied to production and costly to replace.

How it fits the paper: The table refines the mechanism by showing which workers benefit from firm-specific demand rents.

TABLE 10
Idiosyncratic shock pass-through among subgroups of workers, full sample

	Sales elasticity IV		
	Log hourly wage (1)	Log monthly salary (2)	<i>N</i> workers t(3)
Perm. contract	0.109** (0.052)	0.130** (0.054)	355,398
Fixed-term contract	0.181 (0.113)	0.193* (0.112)	69,371
Male	0.080 (0.054)	0.094* (0.055)	199,962
Female	0.118* (0.069)	0.151** (0.077)	155,436
Low wage	0.139** (0.061)	0.153** (0.063)	217,981
High wage	0.090* (0.052)	0.114* (0.054)	212,722
No high school	0.143** (0.064)	0.169** (0.068)	267,194
High school	0.025 (0.056)	0.034 (0.057)	109,153
Blue collar	0.147** (0.061)	0.169** (0.064)	220,827
White collar	0.027 (0.062)	0.041 (0.064)	119,265
BC lower-skilled	0.054 (0.065)	0.072 (0.064)	88,698
BC semi-skilled	0.195** (0.078)	0.221** (0.084)	132,129

Notes: Table displays idiosyncratic shock pass-through elasticities corresponding to IV estimates from columns 1 and 2 of Table 7, pertaining to specific subgroups of workers. Wage outcomes are defined as within-spell, and observations when workers are not employed by their 2007 employer are treated as missing. The sample is split into high and low friction subsamples based on whether each of three different measures of potential frictions is above or below median for the sample. Each estimate is obtained from a separate equation. Specification is identical to that in Table 7. “High-wage” and “Low-wage” indicate 2007 wage above/below 2007 firm median (taken over all workers, not just attached). “Blue Collar” and “White Collar” positions are distinguished based on the occupational coding of the job. We code craftman and foreman jobs as “Semi-Skilled” blue-collar (BC) work and operator and labourer jobs as “Lower-Skill” blue-collar jobs. Number of firms and incumbent workers included in each specification are displayed. All regressions include year fixed effects, as well as controls for year-specific effects of the share of exports going to Spain or Angola in 2005–7. When the $\Delta_{j(t)}^{id}$ is the shock, we include a control for $\Delta_{j(t)}^{com}$. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Idiosyncratic shock pass-through among subgroups of workers.

6 Short summary

- The paper separates firm-specific demand shocks from market-wide demand shocks.
- Idiosyncratic shocks affect own firms but not close competitors.
- Idiosyncratic shocks pass through to incumbent wages with elasticities around 0.10–0.15.
- Common shocks affect both own and competitor wages, consistent with labour-market competition.
- Pass-through is strongest in low-turnover sectors, supporting the replacement-cost mechanism.

7 Conclusion

7.1 Core conclusion

Garin and Silvério show that firm-specific product demand has causal wage effects for incumbent workers. This means wages are partly set inside durable employment relationships rather than only by market-wide competitive wages.

7.2 Mechanism

The key mechanism is demand-sensitive replacement cost. Workers gain bargaining power when the firm has more to lose from worker departure. This mechanism explains why idiosyncratic shocks matter in high-friction sectors.

7.3 Common versus idiosyncratic shocks

Common shocks affect outside options and rival employers, while idiosyncratic shocks isolate internal rent sharing. The distinction is the paper's central empirical contribution.

7.4 External validity

The setting is Portugal during a recession with strong employment protection. Effects may be smaller in more fluid labour markets, but the heterogeneity analysis shows exactly how fluidity changes wage incidence.

7.5 Takeaway

Employer demand and labour-market competition both matter. Which channel dominates depends on turnover costs and relationship durability.